

CRISTAL Lab Research Statement Guide

(PhD and Postdoctoral Applications)

The CRISTAL Lab at CISPA Helmholtz Center for Information Security studies trustworthy AI systems, with a focus on alignment, safety, reasoning, and transparency.

This guide is intended to help applicants write strong research statements and to clarify what we look for during evaluation.

Purpose of the Research Statement

The research statement is one of the most important parts of your application. It allows us to understand:

- your scientific thinking and problem formulation ability
- your research trajectory and evolution over time
- your ability to propose meaningful future directions
- your alignment with CRISTAL's research themes

We value clear scientific thinking, original ideas, and well-thought-out writing over exhaustive summaries.

Recommended Structure (2–3 pages)

We strongly recommend structuring your statement as follows:

1. Research Background (approximately 0.5–1 page)

Briefly describe your past research experience.

Focus on:

- What research problems you worked on
- What your specific contributions were
- What methods or systems you used or developed
- What you learned scientifically from this work

We value clarity over breadth. It is better to explain a few contributions well than to list many projects superficially.

2. Research Vision (approximately 1–1.5 pages)

Describe the research directions you would like to pursue next.

This is the most important section.

We encourage you to address:

- What research problems do you find most important and why?
- What critical limitations do you see in current state-of-the-art approaches?
- What new ideas, hypotheses, or research directions you would like to explore?
- How would you practically begin to investigate these questions?

Note: You do not need to provide a fully developed research plan, but we value specific and thoughtful scientific direction, not general statements.

3. Fit to CRISTAL (approximately 0.5 page)

Briefly explain how your research interests connect to CRISTAL's themes:

- AI Alignment
- Safety Evaluation
- Transparent Decision-Making
- Reasoning and robust AI systems
- Agentic AI

Tell us where you see a natural bridge between your prior work, unique perspective, or skills and CRISTAL's goals. We want to see how your vision could deepen one of our core directions, open a complementary line of inquiry, or spark exciting new collaborations within the lab.

What We Look For

Strong research statements typically demonstrate:

- Clear problem formulation skills
- Thoughtful, articulate, and well-structured scientific writing
- Ability to reason about system failures and limitations
- Originality in thinking about research directions
- Depth of understanding in at least one area
- Scientific curiosity and independence

What We Do NOT Expect

We do not expect:

- A complete research proposal
- An exhaustive literature survey
- Coverage of all CRISTAL research areas

Common Pitfalls

Some common issues that weaken research statements:

- Rewriting your CV in paragraph form
- Listing projects without explaining your role or insights
- Writing overly broad statements ("I am interested in AI safety and LLMs")
- Describing many topics without depth
- Focusing only on past work without future direction

What Strong Statements Often Look Like

Strong statements often include:

- One or two clearly defined research threads
- Explicit questions or hypotheses
- Reflection on limitations of existing methods
- A sense of direction and intellectual curiosity
- Evidence of independent thinking
- Explicit connections between your research vision and submitted artifacts (e.g., "In my repository [X], I encountered limitation [Y], which inspires my proposed future direction to...").

Final Note

We view the research statement as a window into how you think as a researcher.

We are excited to read statements that reflect curiosity, depth, and a genuine interest in building trustworthy and human-aligned AI systems.